

An Agentic LLM Copilot for Real-Time Aircraft Turnaround Monitoring and Delay-Recovery Recommendation

Reasoning over A-CDM milestone event streams to predict off-block breaches and recommend recovery actions

1. Introduction

Aircraft turnaround — the ground phase between an aircraft's arrival on stand and its departure — is one of the most time-critical and tightly coupled processes in aviation operations. A short-haul, low-cost turnaround is frequently completed in 25–35 minutes and requires the parallel and sequenced coordination of disembarkation, cleaning, fuelling, catering, loading, boarding and pushback, each subject to strict dependencies and time windows. The Airport Collaborative Decision Making (A-CDM) framework, jointly developed by EUROCONTROL, ACI EUROPE, IATA and CANSO, formalises this process through a shared set of milestones and target times such as the Target Off-Block Time (TOBT), enabling partners to detect deviations and trigger collaborative re-planning.

Within this setting the turnaround coordinator must continuously fuse incoming milestone events, anticipate whether the aircraft will meet its TOBT, and decide which recovery actions to take when a milestone slips. This is fundamentally an agentic decision problem: perceive state, predict outcomes, plan, act and re-plan. Recent advances in agentic Large Language Models (LLMs) — models augmented with planning, tool use and reflection — make it newly feasible to provide intelligent decision support for exactly this kind of dynamic, constraint-rich coordination task.

This project has been chosen because it sits at the intersection of a genuine, costly industry problem and an active, under-explored research frontier. Reactionary (knock-on) delay accounts for a substantial share of total departure delay and propagates through aircraft rotations across the network. Yet the existing turnaround-delay literature is dominated by classical machine-learning forecasting, while agentic LLMs have not been meaningfully applied to live turnaround coordination. The combination of the candidate's operational ground-handling experience and the maturing agentic-LLM toolkit creates an opportunity to investigate whether such a system can add value beyond both classical schedulers and human coordinators working unaided.

2. Problem Statement

The principal problem is that turnaround coordinators must make rapid, high-stakes sequencing and recovery decisions under uncertainty, using fragmented and continuously arriving information, with little decision-support tooling beyond status dashboards. A dashboard reports that a milestone is late; it does not reason about what that lateness will cause, nor recommend what to do about it. The coordinator must perform this causal and recovery reasoning mentally, in real time, often across several concurrent aircraft.

Concretely, the problem decomposes into three coupled sub-problems. First, detection and prediction: given the stream of milestone timestamps observed so far, will the aircraft meet its TOBT, and if not, by how much? Second, propagation reasoning: a delayed milestone rarely affects only itself — because turnaround tasks are partly sequential (for example boarding cannot complete before fuelling concludes under with-passengers fuelling constraints), a slip propagates through the dependency structure of the turnaround. Third, recovery: given a predicted breach, which actions (expediting a service, re-sequencing tasks, re-allocating a team, or requesting a TOBT update) best protect the departure, and how should they be communicated to the relevant stakeholders?

Those affected by the problem are several and aligned in their interest in punctuality. Ground-handling providers are measured on on-time performance and milestone adherence and bear contractual exposure when targets are missed. Airlines incur direct and reactionary delay costs and reputational harm. Airports and air navigation service providers lose slot and stand predictability. Passengers experience the downstream disruption. Because reactionary delay propagates along aircraft rotations, a single poorly recovered turnaround can degrade punctuality far beyond the originating flight.

Solving the problem matters because even modest improvements in early breach detection and recovery decision quality compound across a high-frequency operation. The intellectual difficulty — and the reason a dashboard or a purely classical scheduler is insufficient — lies in reasoning over messy, incomplete, and partly unmodelled real-world events (for example a verbally reported jammed cargo door), generating context-specific and explainable recommendations, and producing the stakeholder communication that turns a decision into coordinated action. These are precisely the capabilities that distinguish an agentic LLM from both a static dashboard and a fixed optimisation model.

The research questions this project will address are:

- RQ1. Can an agentic LLM, reasoning over a live A-CDM milestone event stream, predict off-block time breaches earlier or more accurately than a rule-based baseline?

- RQ2. Does equipping the agent with tools — including a critical-path / scheduling solver — improve the quality of its recovery recommendations relative to the LLM reasoning unaided?
- RQ3. How does the agentic approach compare against a classical scheduling baseline and an unaided human, in recovery decision quality, explainability, and timeliness, and where does it fail?

3. Aims and Objectives

The overall aim of the project is to design, implement and evaluate an agentic LLM “copilot” for the turnaround coordinator that ingests a stream of turnaround milestone events, predicts off-block breaches, reasons about delay propagation, and recommends explainable recovery actions — and to characterise rigorously where this approach adds value over established baselines and where it does not. The system is deliberately positioned as decision support: the agent recommends and the human decides, reflecting the safety-critical nature of aviation operations and making the contribution both more credible and more ethically defensible.

The work will be guided by the following objectives:

- O1. **Domain model and simulator.** Build a faithful, configurable simulation of a short-haul turnaround as a dependency graph of tasks with durations, precedence constraints and resource limits, grounded in the A-CDM milestone framework, capable of generating realistic milestone event streams and injecting controlled disruptions (late inbound, fuelling delay, crewing shortfall, equipment fault).
- O2. **Baselines.** Implement a rule-based / threshold coordinator and a classical critical-path (scheduling) baseline against which the agent will be measured.
- O3. **Agent architecture.** Design an agentic LLM that operates a perceive–predict–plan–act–reflect loop over the event stream, exposing tools such as a milestone-status query, a delay-propagation estimator, the critical-path solver, and a stakeholder-communication drafter.
- O4. **Recovery and communication.** Enable the agent to generate ranked, explainable recovery recommendations and accompanying draft communications for relevant stakeholders.
- O5. **Evaluation.** Evaluate the agent against the baselines across many seeded, disrupted turnarounds, and conduct a small expert assessment of recommendation quality.
- O6. **Analysis.** Critically analyse the results, including failure modes, latency, and the marginal value of the agentic layer over classical scheduling.

The approach to the principal problem is a simulation-based experimental study. Because comprehensive operational data is severely restricted in aviation owing to commercial sensitivity, a configurable simulator is the appropriate and now well-precedented methodology: it provides ground truth (actual off-block outcome and milestone adherence), permits controlled and repeatable injection of disruptions, and allows rigorous comparison against baselines. The agent will be implemented using a current LLM with a structured tool-use / function-calling loop; the critical-path solver and propagation estimator will be conventional algorithmic tools the agent can invoke, embodying the “LLM as orchestrator” pattern. Evaluation will be primarily quantitative — on-time performance, milestone adherence, prediction lead time and error, and recovery effectiveness versus baselines — complemented by a qualitative expert rating of recommendation quality and explanation usefulness. The research strategy is therefore a constructive (design-and-build) study validated through controlled experimentation, which is well suited to an applied AI systems project of this kind.

4. Legal, Social, Ethical and Professional Considerations

Legal and commercial. The project will use no proprietary or commercially sensitive operational data. The setting will be described generically (an integrator providing ground handling for a low-cost carrier at a UK regional airport) and the simulator will be parameterised from publicly available A-CDM specifications and published literature rather than from any employer's internal procedures or contracts. Any use of real organisational context will be deferred pending a confidentiality and contract review, and the proposal is written to remain valid without it.

Safety and professional responsibility. Aviation ground operations are safety-related. The system is therefore designed strictly as decision support: it recommends, and a qualified human retains authority for every operational decision. The dissertation will be explicit that the artefact is a research prototype evaluated in simulation, not a deployable operational tool, and will discuss the assurance, validation and human-factors barriers that would precede any real-world use. This aligns with professional codes (for example the BCS Code of Conduct) emphasising public safety and competence.

Social and human factors. Automation in coordination roles raises concerns about deskilling, automation bias and accountability. The framing as a copilot, the emphasis on explainable recommendations, and the inclusion of failure-mode analysis directly address the risk of misplaced trust in LLM output, including the well-documented tendency of LLMs to produce fluent but incorrect (hallucinated) reasoning.

Ethical conduct of the study. Any expert evaluation involving human participants will follow the university's ethics procedures, with informed consent and anonymised, aggregated reporting. No personal data of passengers or staff will be processed. Overall

project risks — principally scope creep and simulator fidelity — will be managed through a staged plan in which a minimal end-to-end pipeline is delivered before optional extensions.

5. Background

5.1 Agentic LLMs: planning, tool use and reflection

The notion of an “agentic” LLM refers to a language model embedded in a loop that perceives an environment, plans, takes actions via tools, observes results and adapts. Surveys of LLM-based autonomous agents organise the field around perception, memory, planning and action, and document a rapid expansion of methods and applications since 2023. Planning in particular has been the subject of dedicated survey work, which taxonomises approaches into task decomposition, plan selection, the use of external modules, reflection, and memory — a framing directly relevant to a turnaround agent that must decompose a recovery problem and select among candidate actions.

A central enabler of agentic behaviour is tool use. Toolformer demonstrated that language models can teach themselves to call external APIs, and subsequent work has surveyed tool learning with foundation models at scale, including retrieval of the right tool from large tool sets and the orchestration of multiple tools across multi-step tasks. Reflection-style methods, in which an agent critiques and revises its own reasoning, have been shown to improve problem-solving performance, while structured planning approaches connect LLMs to classical planners — for example by generating problem specifications for symbolic (PDDL) planners. This last line of work is the conceptual precedent for the architecture proposed here, in which the LLM orchestrates a classical critical-path solver rather than attempting to perform exact scheduling itself.

The field is maturing but far from settled. Recent surveys of agent evaluation and of why multi-agent LLM systems fail emphasise that agentic systems are brittle: they suffer from error propagation across steps, unreliable tool selection, and difficulty recovering from their own mistakes. These known weaknesses motivate both the bounded scope of this project and its central comparative question — whether the agentic layer genuinely adds value over simpler, more predictable methods. They also justify the decision-support framing, since the literature does not support handing autonomous control to such systems in a safety-related domain.

5.2 Aircraft turnaround and A-CDM

The operational backbone for the proposed system is Airport Collaborative Decision Making (A-CDM). A-CDM is a joint initiative of EUROCONTROL, ACI EUROPE, IATA and CANSO that seeks to reduce delays, increase the predictability of events during a flight, and optimise the use of resources by sharing accurate and timely information

among partners. Its core is the Milestones Approach, which describes the progress of a flight from initial planning through to take-off via a sequence of defined milestones; European implementations commonly use a set of sixteen milestones spanning the inbound, turnaround and outbound phases. Passing a milestone triggers updates to downstream time elements and allows deviations to be detected and collaborative decisions to be made.

The pivotal time element for the turnaround is the Target Off-Block Time (TOBT) — the time at which the aircraft operator or ground handler expects the aircraft to be ready, doors closed and ready for pushback — from which the air navigation service provider derives the Target Start-up Approval Time (TSAT) and ultimately the Target Take-Off Time (TTOT). The EUROCONTROL A-CDM specification defines the alerting and update logic around these times, for example raising alerts when TOBT and the filed estimated off-block time diverge beyond defined thresholds, and updating TOBT when it expires. This rule structure is valuable for the project because it provides a documented, industry-standard backbone for the simulator's event model and for the rule-based baseline, rather than an invented one. Industry guidance further stresses the active use of actual milestone timestamps for in-block, boarding and off-block to identify bottlenecks and the early sharing of estimated disruptions — exactly the signal the proposed agent will consume.

It is worth noting that the milestones, while numbered, do not necessarily occur strictly sequentially, and many depend on local procedures. This partial ordering, with concurrency and local variation, is precisely what makes the turnaround a non-trivial dependency-graph problem and what makes flexible reasoning over incomplete event streams valuable.

5.3 Delay prediction and propagation: the state of the art

The existing scholarly treatment of turnaround delay is dominated by classical and statistical machine learning. A well-developed strand predicts flight delays and turnaround times directly: two-stage frameworks first classify whether a flight will be delayed and identify a cause category, then predict duration, using features such as flight timing, airport demand and weather. Probabilistic models have been developed specifically to forecast turnaround times and TOBT at major European airports, providing distributions rather than point estimates and explicitly recognising the turnaround as a critical contributor to reactionary delay.

A second strand models delay propagation along aircraft rotations. Chained prediction approaches link an aircraft's departure delay to the arrival delay of its next flight, building a propagation model that traces delay along an itinerary. Data-driven aircraft-routing work reconstructs primary (root-cause) delays via quantile regression and then predicts them with ensemble methods, in order to build buffers and robustness into schedules. More

recent work integrates a delay-absorption stage, learning the probability that an upstream delay is absorbed during turnaround before the next departure — an explicit, if still classical, treatment of recovery. Across this literature, gradient-boosted trees (XGBoost, LightGBM, CatBoost) and random forests recur as the dominant predictive tools, frequently noting that comprehensive operational data is severely restricted by commercial sensitivity — a constraint that has itself motivated the use of synthetic aviation data.

Two observations follow. First, the classical literature is strong at point or probabilistic prediction of delay magnitude but treats propagation largely as static features and offers little in the way of explainable, action-level recovery recommendation or stakeholder communication. Second, and decisively for this project, agentic LLMs appear essentially absent from the live turnaround-coordination setting. The gap this project addresses is therefore the bridge between two mature but separate fields: agentic LLM planning and tool use on one side, and turnaround / delay-propagation modelling on the other.

5.4 Relationship to this project and the state of the art

Against this backdrop, the proposed work is neither a re-implementation of existing delay predictors nor a purely speculative application. It is an extension that recombines established components in a new configuration: a documented A-CDM milestone model and classical critical-path scheduling (well established) are wrapped as tools inside an agentic LLM loop (well established in general, but novel in this domain) to deliver early breach prediction, propagation-aware reasoning, and explainable recovery recommendations. The techniques are individually mature, which makes the project tractable within an MSc; their combination in the turnaround domain is, to the best of the available evidence, new.

The honest research position is comparative rather than triumphal. Classical optimisation may well match or beat the agent on pure scheduling accuracy; the hypothesised advantage of the agentic approach lies in handling ambiguous and unmodelled events, generating human-readable explanations, and producing communication — and in flexibly invoking the classical solver as one tool among several. By measuring the agent against both a rule-based baseline and a classical scheduler, and by analysing failure modes informed by the known brittleness of agentic systems, the project aims to produce a credible, defensible characterisation of where agentic LLMs help turnaround coordination and where they do not.

The results are likely to interest parties beyond the university. Ground-handling integrators, low-cost carriers, airports and ANSPs all have a direct operational and financial stake in turnaround punctuality and reactionary-delay containment, and there is active commercial and research interest in turnaround monitoring and A-CDM tooling. A

rigorous, even partly negative, finding about the value of agentic LLMs in this setting would be informative to that audience.

6. References

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Student and First Supervisor Project Sign-Off

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SUPERVISOR: I approve this project proposal:			
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